Does Money Buy National Happiness? It Depends on Who You Ask and How*

Adrien Fabre†

CNRS, CIRED, France

Abstract

Is GDP per capita a good predictor of national well-being? I compare it with a simple alternative, a country's broad world region, in the two main sources of cross-national well-being data. In the World Values Survey and the European Values Study (WVS, 1981–2023, 112 countries), log GDP per capita explains on average 14% of the cross-country variance of eight well-being indicators, against 33% for region, which is the better predictor, in and out of sample, in 88% of 952 specifications; this advantage stems from Latin America and Eastern Europe. In the Gallup World Poll (2006–2023, 168 countries), by contrast, income predicts life evaluations as well as region. To understand why the two datasets disagree, I randomize the question wording (Gallup's Cantril ladder vs. WVS's life satisfaction) and scale (0–10 vs. 1–10) in an online survey of 11,000 respondents in ten high-income countries. Asked to the same sample, the ladder is better predicted by GDP than life satisfaction (R^2 of 0.61 vs. 0.28 across countries), which reproduces about half of the Gallup–WVS difference in predictive capacity, but the estimated income gradients of both questions are as steep as Gallup's, three times steeper than the WVS's. Wording and samples thus both contribute to the discrepancy, and the new data lie midway between Gallup and the WVS.

Keywords: subjective well-being; life satisfaction; GDP; world regions; survey methodology; question wording.
JEL codes: I31; O15; O57; C83.

Data availability: the code, the data of the survey experiment, the GDP and other World Bank and IMF series, and the country-level well-being indicators computed from the WVS, EVS and Gallup data are publicly available at https://github.com/bixiou/wellbeing_gdp_region, with instructions to reproduce all results. The third-party data can be obtained as follows. The WVS time-series file (version 3.0.0) is freely available, upon registration, at https://doi.org/10.14281/18241.23, and the Joint EVS/WVS 2017–2022 dataset (ZA7505, version 5.0.0) from GESIS at https://doi.org/10.4232/1.14320. The World Happiness Report data are freely available at https://worldhappiness.report/data-sharing/. World Bank data are freely available at https://databank.worldbank.org/source/world-development-indicators and https://pip.worldbank.org, and IMF data at https://www.imf.org/external/datamapper. The Gallup World Poll microdata are proprietary and must be licensed from Gallup (https://www.gallup.com/analytics/318875/global-research.aspx); the distributions of answers to the Cantril ladder by country and wave used in this paper are included in the repository. *Acknowledgments:* I thank Claudia Senik, whose 2016 course assignment was the starting point of this paper, participants in the 2024 STATEC Well-being Conference for helpful comments, Armon Rezai and the Vienna University of Economics and Business (WU Wien) for access to the Gallup World Poll data, and Julieta Toffoli for research assistance. *Funding:* this research received funding from Agence Nationale de la Recherche (ANR-24-CE03-7110). *Declaration of interest:* none. *Declaration of generative AI and AI-assisted technologies in the writing process:* during the preparation of this work the author used Claude Opus 5.5 to write analysis code and draft parts of the manuscript. After using this tool, the author reviewed and edited the code and content created by AI as needed and takes full responsibility for the content of the publication.

†Corresponding author. CNRS, CIRED, 45 bis avenue de la Belle Gabrielle, 94736 Nogent-sur-Marne Cedex, France. E-mail: adrien.fabre@cnrs.fr. ORCID: 0000-0001-5245-7772.

1 Introduction

Which country is the happiest? The answer most often heard, based on the World Happiness Report, is a Scandinavian, high-income country. Behind this answer lies one of the most robust stylized facts of well-being research: across countries, average life evaluations increase with the logarithm of GDP per capita (Deaton, 2008; Sacks et al., 2012; Stevenson & Wolfers, 2008). Using the Gallup World Poll, Deaton (2008) finds a correlation above 0.8 between average life evaluation and log GDP per capita; using the World Values Survey (WVS), Inglehart & Klingemann (2000) reported an R^2 of about one half for an index of happiness and satisfaction. These correlations are frequently invoked to support the view that economic growth is a reliable route to higher well-being.

This paper revisits the predictive capacity of GDP per capita for national well-being, and asks two questions. First, how much of the cross-country variation in well-being does GDP per capita actually predict, and does it predict it better than a very simple alternative, the country's world region? Second, why do the two main sources of cross-national well-being data, Gallup and the WVS, give different answers to the first question? Previous work has noted that income is more strongly related to Gallup's ladder than to WVS measures, and hypothesized that part of the difference was due to a lack of representativeness of WVS samples in developing countries, which were "mostly literate and urban" (Deaton, 2008); it has also shown that national income is more related to life evaluations than to happiness or emotions (Inglehart et al., 2008; Kahneman & Deaton, 2010). Yet whether the difference between datasets comes from the question asked or from the samples has, to my knowledge, not been tested experimentally.

For the first question, I use the two main sources of cross-national well-being data: all seven waves of the WVS, complemented with the 2017 wave of the European Values Study (342 country-years, 112 countries), and the Gallup World Poll (2,351 country-years, 168 countries). With eight indicators of national well-being in the WVS, seven in Gallup, and seven income measures, I compare the variance explained by income and by region using the Shapley (LMG) decomposition of R^2 (Grömping, 2007; Lindeman et al., 1980) and leave-one-country-out cross-validation. For the second question, I embedded a 2×2 experiment in an online survey of 11,000 respondents in ten high-income countries, randomizing the wording (Gallup's Cantril ladder vs. the WVS life-satisfaction question) and the scale (0–10 vs. 1–10). Combining the experiment with Gallup and WVS data for the same countries, I decompose the Gallup–WVS gap into a question effect and a residual reflecting differences in samples, modes, context and time.

The results are threefold. First, the two datasets disagree. In the WVS, GDP per capita

is a weak predictor of national well-being: on average over the eight indicators, it explains 14% of the cross-country variance (95% CI [9%; 20%]), against 33% for region ([26%; 42%]), and it explains nothing of the share of very happy people. Region is the better predictor in 88% of specifications and, out of sample, for every indicator. In Gallup, by contrast, log GDP per capita explains 61% of the cross-country variance of the mean ladder ([52%; 69%]), against 48% for region ([38%; 61%]). Second, the region advantage in the WVS rests on two regional deviations from the income gradient: Latin America, happier than its income predicts, and Eastern Europe, unhappier. Third, the experiment shows that the question explains part, but not all, of the discrepancy between the two datasets. In the ten countries, log GDP explains 0.75 of the variance of Gallup's ladder and 0.16 of WVS satisfaction. Asked to the same online sample, the ladder is also better predicted by GDP than the satisfaction question (R^2 of 0.61 vs. 0.28). Wording accounts for about half (56%) of the Gallup–WVS difference in predictive capacity, and the new data lie midway between the two datasets. However, the estimated income gradient is as steep as Gallup's for both questions, although imprecisely estimated with ten countries (slopes of 3.60 and 3.86, vs. 4.20 in Gallup and 1.36 in the WVS): the satisfaction question adds dispersion around the income gradient, but the flatter WVS gradient appears to be specific to its samples. Wording also has a sizeable effect on levels: the ladder shifts answers by -0.45 points, which explains why Gallup levels are lower than WVS levels in high-income countries.

The paper contributes to three literatures. It adds to the debate on the relationship between income and well-being across countries and over time (Deaton, 2008; Easterlin, 1974; Easterlin et al., 2010; Jebb et al., 2018; Pirla et al., 2025; Powdthavee et al., 2017; Proto & Rustichini, 2015; Stevenson & Wolfers, 2008) by showing that the strength of the cross-country relationship depends on the dataset, and by benchmarking income against regions. It relates to work on regional and cultural differences in well-being, in particular the high well-being of Latin America (Graham & Lora, 2009; Rojas, 2016) and the low well-being of transition countries (Guriev & Zhuravskaya, 2009), and on the cultural foundations and cross-cultural comparability of well-being (Conzo et al., 2017; Diener & Suh, 2000; Exton et al., 2015; Senik, 2014). Finally, it contributes to the methodological literature on the measurement of subjective well-being (Bond & Lang, 2019; Kahneman & Krueger, 2006; Kaiser & Oswald, 2022; OECD, 2013; Stone & Mackie, 2013), and specifically on the comparability of the Cantril ladder and life-satisfaction questions (Bjørnskov, 2010; Nilsson et al., 2024) and on context and mode effects (Deaton & Stone, 2016; Dolan & Kavetsos, 2016), by providing experimental estimates of wording and scale effects in ten countries. In this respect, it echoes evidence that life evaluations and affective well-being are distinct, if correlated, dimensions of well-being (Berlin & Fors Connolly, 2019).

Section 2 presents the data. Section 3 compares income and region as predictors of national well-being. Section 4 investigates the discrepancy between Gallup and the WVS. Section 5 discusses limitations, and Section 6 concludes.

2 Data

2.1 World Values Survey

The World Values Survey (WVS) is the longest-running cross-national survey of values and subjective well-being. I use the WVS time-series file (Haerpfer et al., 2022), which pools the seven waves conducted between 1981 and 2022, and complete it with the Joint EVS/WVS 2017–2022 dataset (EVS/WVS, 2024), which adds the 36 national surveys of the 2017 wave of the European Values Study (EVS, conducted in 2017–2021) and 2 recent WVS surveys (India 2023, Uzbekistan 2022). The EVS asks the same well-being questions, and the combination of the two surveys is standard practice (the so-called Integrated Values Surveys). The EVS and the WVS are distinct surveys: when both surveyed the same country the same year (3 cases), I keep both surveys as separate observations. For brevity, I refer to the combined data as the WVS: 501,911 respondents in 342 country-years and 112 countries. National samples are designed to be representative of the adult population, typically with 1,000 to 3,000 respondents, and most interviews are conducted face-to-face (though some recent national surveys used web, mail or mixed modes, e.g. the WVS in the United States in 2017 and in Japan in 2019, or the EVS in Switzerland in 2017). All statistics use the survey weights provided by the WVS and the EVS.

The WVS contains two questions on subjective well-being:

- *Happiness*: "Taking all things together, would you say you are: *Very happy*; *Quite happy*; *Not very happy*; *Not at all happy*."
- *Life satisfaction*: "All things considered, how satisfied are you with your life as a whole these days?", answered on a scale from 1 (*Completely dissatisfied*) to 10 (*Completely satisfied*).

Non-responses are rare (1.4% for satisfaction and 1.7% for happiness on average) and uncorrelated with GDP per capita (correlation: -0.09); they are excluded.

Indicators of national well-being. Because there is no consensus on how to aggregate ordinal answers into a national indicator, and because the ranking of countries can de-

pend on this choice (Bond & Lang, 2019), I compute eight indicators for each country-year:

- *Happy*: share answering *Quite* or *Very happy*;
- *Very Happy*: share answering *Very happy*;
- *Very Unhappy*: share answering *Not at all happy*;
- *V. Happy – V. Unhappy*: difference between the two previous shares;
- *Happiness (mean)*: mean happiness, with answers coded $+3$, $+1$, -1, -3;
- *Satisfaction (mean)*: mean life satisfaction;
- *Satisfied*: share answering 6 to 10 to the life-satisfaction question;
- *Happy + Satisfied*: average of *Happy* and *Satisfied*, the index used by Inglehart & Klingemann (2000).

The indicators are positively but imperfectly correlated: across country-years, the correlation between *Happy* and *Satisfied* is 0.72, and between *Very Happy* and *Satisfaction (mean)* it is only 0.60 (Online Appendix Table A1). For the life-satisfaction question, I also compute the shares answering 8–10, 9–10, 10, 1–4 and 1–2, for comparisons with Gallup.

2.2 Gallup World Poll and World Happiness Report

The Gallup World Poll interviews around 1,000 adults per country and year in more than 140 countries, by telephone where coverage is high (typically in high-income countries) and face-to-face elsewhere (Gallup, 2022). Its life-evaluation question is the Cantril ladder (Cantril, 1965): "Please imagine a ladder, with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best possible life for you and the bottom of the ladder represents the worst possible life for you. On which step of the ladder would you say you personally feel you stand at this time?"

I use the distribution of answers by country and wave, tabulated from the Gallup World Poll microdata (2,351 country-years, 168 countries, 2006–2023), from which I compute the mean and the shares answering 6–10, 8–10, 9–10, 10, 0–4 and 0–2. These distributions are unweighted. To cover the most recent years, I also use the ladder means published in the World Happiness Report 2026 (World Happiness Report, 2026), which average Gallup data over three years up to 2023–2025.

2.3 Income

My benchmark income measure is the log of GDP per capita at purchasing power parity (PPP, constant 2021 international dollars) from the World Development Indicators (World Bank, 2026b). As an alternative, I use GDP per capita at market exchange rates (constant 2015 US dollars). Missing values are imputed in three steps. First, PPP series, which start in 1990, are extended backwards (and gaps filled) using the growth rate of real GDP per capita. Second, for countries absent from the World Bank data (e.g. Taiwan, or Venezuela in recent years), I use the IMF World Economic Outlook (International Monetary Fund, 2026), converted into the constant dollars of the World Bank series with a U.S. price deflator (the ratio of World Bank to IMF values for the United States). Third, remaining gaps are filled with the closest year available (Montenegro 1996). Territories without national accounts use the GDP of the closest entity (e.g. the United Kingdom for Northern Ireland). A robustness check excludes all imputed values.

Region is a categorical variable, which can capture non-linear patterns that a single continuous variable cannot. To compare income and region on an equal footing, I also use categorical income indicators: income sextiles and income clusters obtained by k-means clustering of log GDP per capita, with $k = 5$, 6 or 7, recomputed in each estimation sample. The clusters let the data choose the thresholds between income groups and allow for non-linear relationships.

2.4 Regions

My main classification groups countries into the five UN regional groups (Figure 1): *Africa*, *Asia* (the Asia-Pacific group, which includes the Middle East and Central Asia), *Eastern Europe* (the former Eastern Bloc, excluding Central Asia), *Latin America* (including the Caribbean), and *Western* countries (Western Europe, North America, Australia, New Zealand, and Turkey). Territories that are not UN members (e.g. Taiwan, Hong Kong, Puerto Rico, Kosovo) are assigned to the group of their neighbors. It departs from the UN groups for two countries only: Israel, a member of the Western group since 2000 because it was denied access to the Asia-Pacific group, is classified in Asia, and Cyprus, a member of the Asia-Pacific group, is classified with Western countries.

For robustness, I use four alternative classifications: the exact UN regional groups (with Israel in the Western group and Cyprus in Asia); a six-region variant separating the Middle East and grouping Central Asia with Eastern Europe; the seven World Bank regions (East Asia and Pacific, Europe and Central Asia, Latin America and the Caribbean, Middle East and North Africa, North America, South Asia, Sub-Saharan Africa); and six

continents (Africa, North America including Central America and the Caribbean, South America, Asia, Europe, Oceania). The World Bank regions isolate Latin America but group Eastern and Western Europe; the continents isolate neither.

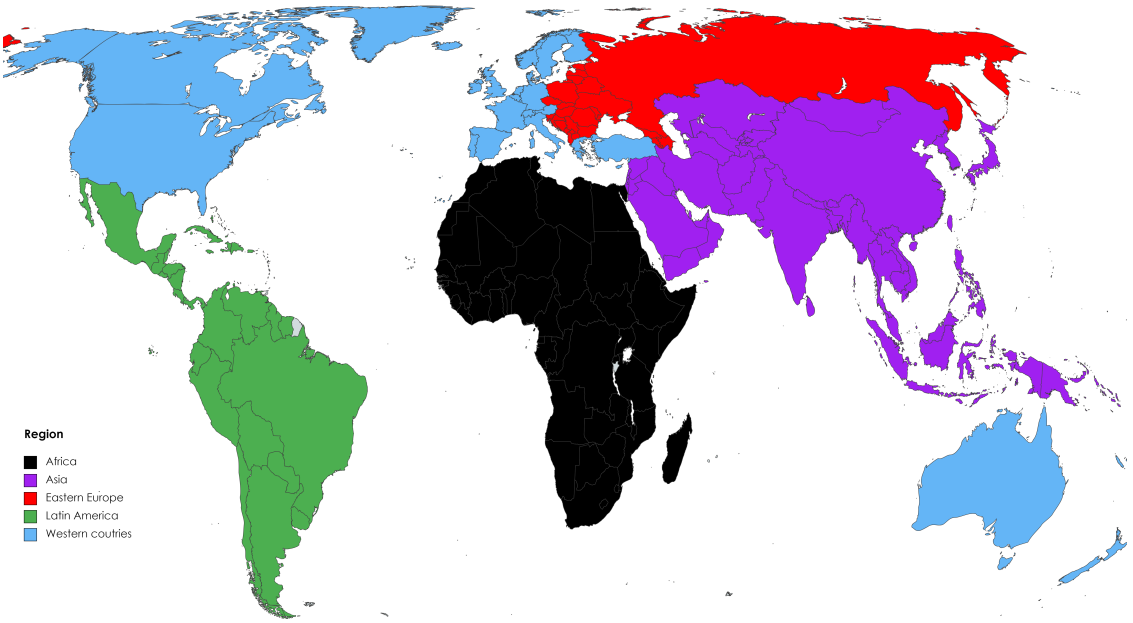


Figure 1: Main classification of countries into five world regions.

2.5 A survey experiment on question wording and scale

Gallup and the WVS differ in many respects. Gallup asks the Cantril ladder on a 0–10 scale, the WVS life satisfaction on a 1–10 scale. Gallup relies on a centralized, standardized design, with about 1,000 respondents per country every year, interviewed by telephone in high-income countries and face-to-face elsewhere; the WVS is fielded every five years or so by national teams, mostly face-to-face, with sampling designs, modes and survey years that vary across countries. These differences fall into two categories that could explain why the two datasets disagree: the question (wording and scale) and everything else—sampling frames, interview modes, questionnaire context, translations, and survey years, which I refer to as *samples* in a broad sense. To separate the two, I embedded a randomized experiment in an original survey (Fabre, 2025). The survey was conducted online between April 15 and July 3, 2025, by Bilendi (or Kantar in Saudi Arabia) on 11,000 respondents in 10 high-income countries: the United States (3,000), Japan (2,000), Saudi Arabia (1,000), Germany (1,048), the United Kingdom (826), France (798), Italy (756), Spain (603), Poland (500) and Switzerland (469). The survey also covered Russia, where the well-being question was censored. Samples were stratified with quotas

on gender, age, income, education, region and urbanicity (as well as race in the U.S. and citizenship in Saudi Arabia), and results are reweighted to match population frequencies along these dimensions (with weights trimmed between 0.25 and 4). The survey was approved by the CIRED institutional review board (IRB-CIRED-2025-2), and it was pre-registered on OSF (https://osf.io/7mzn4). Its main purpose was to study attitudes toward global policies, which are analyzed in a separate paper (Fabre, 2025); the life-evaluation question and its four variants described below are the only part of the survey used here. The sample sizes were set for the main purpose of the survey, not on the basis of a power analysis for the present experiment; I therefore report confidence intervals throughout, and I do not interpret non-significant estimates as evidence of the absence of an effect.

Toward the end of the questionnaire, after questions on global policies, each respondent was randomly assigned to one of four versions of the life-evaluation question:

- *Ladder 0–10* (Gallup's original): the Cantril ladder question quoted in Section 2.2, with steps numbered from 0 (*Worst possible*) to 10 (*Best possible*);
- *Ladder 1–10*: same wording, with steps numbered from 1 to 10;
- *Satisfaction 0–10*: the WVS question quoted in Section 2.1, with answers from 0 (*Completely dissatisfied*) to 10 (*Completely satisfied*);
- *Satisfaction 1–10* (WVS's original): same wording, with answers from 1 to 10.

This 2×2 design identifies the effect of wording (ladder vs. satisfaction) and scale (0–10 vs. 1–10), and their interaction, holding the sample, mode, context and time constant. Within each country, each branch receives around 25% of respondents. The exact wording of the question and of the answer labels is reproduced in Online Appendix OA.1.

3 Income and region as predictors of national well-being

3.1 Empirical strategy

For each well-being indicator and each income indicator, I estimate three linear regressions across country-years i:

$$well\text{-}being_i = \alpha_1 + \beta_1\, income_i + u_i \tag{1}$$

$$well\text{-}being_i = \alpha_2 + \gamma_2\, region_i + e_i \tag{2}$$

$$well\text{-}being_i = \alpha_3 + \beta_3\, income_i + \gamma_3\, region_i + \varepsilon_i \tag{3}$$

where $region_i$ denotes the vector of region dummies. Denoting R^2_1, R^2_2 and R^2_3 their coefficients of determination, the share of the variance explained by income alone is R^2_1. To compare income and region, I decompose R^2_3 between the two regressors following Lindeman et al. (1980) (LMG), i.e. using the Shapley value (Grömping, 2007). With two (groups of) regressors, the LMG contribution of income is the average of its contribution when entered first (R^2_1) and when entered second ($R^2_3 - R^2_2$). The *share of explained variance due to income* is therefore

$$s = \frac{R^2_1 + (R^2_3 - R^2_2)}{2\,R^2_3}, \tag{4}$$

and region is the better predictor whenever $s < 50\%$.

One feature of this comparison could favor region: region has four degrees of freedom while log GDP has one. I address this concern in three ways. First, I use categorical income variables (sextiles and clusters) with as many or more categories as regions. Second, I compute s with adjusted R^2, which penalizes additional regressors; with more than 300 observations, the adjustment is small and only reinforces the conclusions. Third, and more directly relevant to *predictive* capacity, I compute leave-one-country-out cross-validated R^2: each country's observations are predicted by a model estimated on all other countries. Unlike the in-sample R^2, the cross-validated R^2 decreases when a model has superfluous parameters, and it is negative when a model predicts worse than the sample mean. Confidence intervals for the main statistics come from 1,000 bootstrap samples of countries (each drawn with all its country-years), which accounts for the repeated observations of the same countries. Other robustness checks include testing each WVS wave separately, excluding the EVS surveys, weighting countries by their population, keeping only the last observation of each country, excluding the pandemic years 2020–2021, excluding imputed GDP values, using the previous vintage of GDP data (PPP in constant 2017 dollars), alternative region classifications, and excluding Latin America or Eastern Europe.

3.2 Income explains a small share of national well-being in the WVS

Figure 2 plots mean life satisfaction against GDP per capita for all WVS country-years. The relationship is positive, but the dispersion at any given income level is large: countries with a GDP per capita around 10,000 dollars span most of the range of observed life satisfaction, from the former Eastern Bloc at the bottom to Latin American countries at the top. Log GDP per capita explains 23% of the variance of mean satisfaction. The relation is even weaker for happiness (Figure 3). Online Appendix Figures A1–A7 plot the other

indicators.

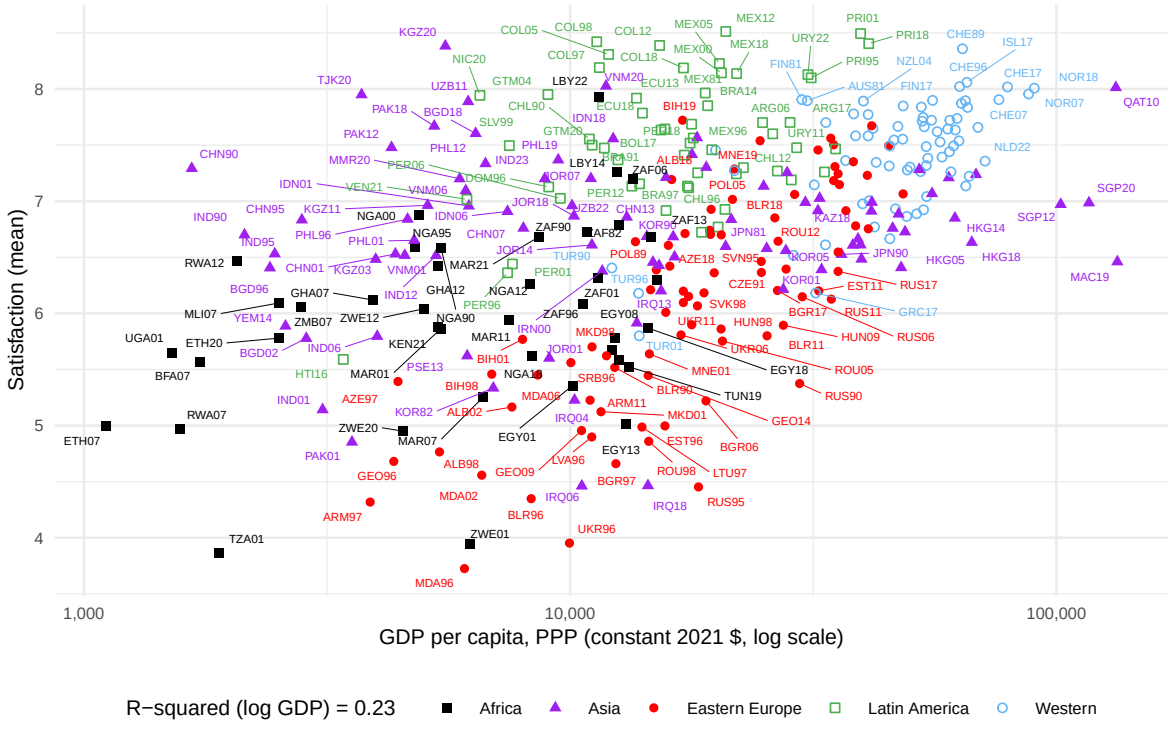


Figure 2: Mean life satisfaction vs. GDP per capita (PPP), WVS/EVS 1981–2023.

Table 1 generalizes this result to all indicators and income measures. On average over the eight indicators, log GDP per capita at PPP explains 14% of the variance (95% CI [9%; 20%]), and the best-performing income measure (Cluster k=7 nominal) explains 20%. Satisfaction-based indicators are better explained than happiness-based ones: log GDP explains 4% of the variance of mean happiness and essentially none of the share of *Very happy* people, whose slope with respect to log GDP is not significantly different from zero (slope 0.002, $p = .948$; Online Appendix Table A2). This confirms the observation of Deaton (2008) and Inglehart et al. (2008) that income is more closely related to life evaluation than to happiness. By contrast, region alone explains 33% of the variance on average ([26%; 42%]), and scores high even on happiness-based variables (last column).

3.3 Region predicts national well-being better than income in the WVS

Table 2 reports the share of explained variance due to income, s, for each indicator and income measure. It is below 50% in all cases but one, where income and region are tied (0.50). On average, log GDP per capita accounts for 23% of the variance explained jointly by income and region (95% CI [16%; 32%]), and the best income measure for 33%. The

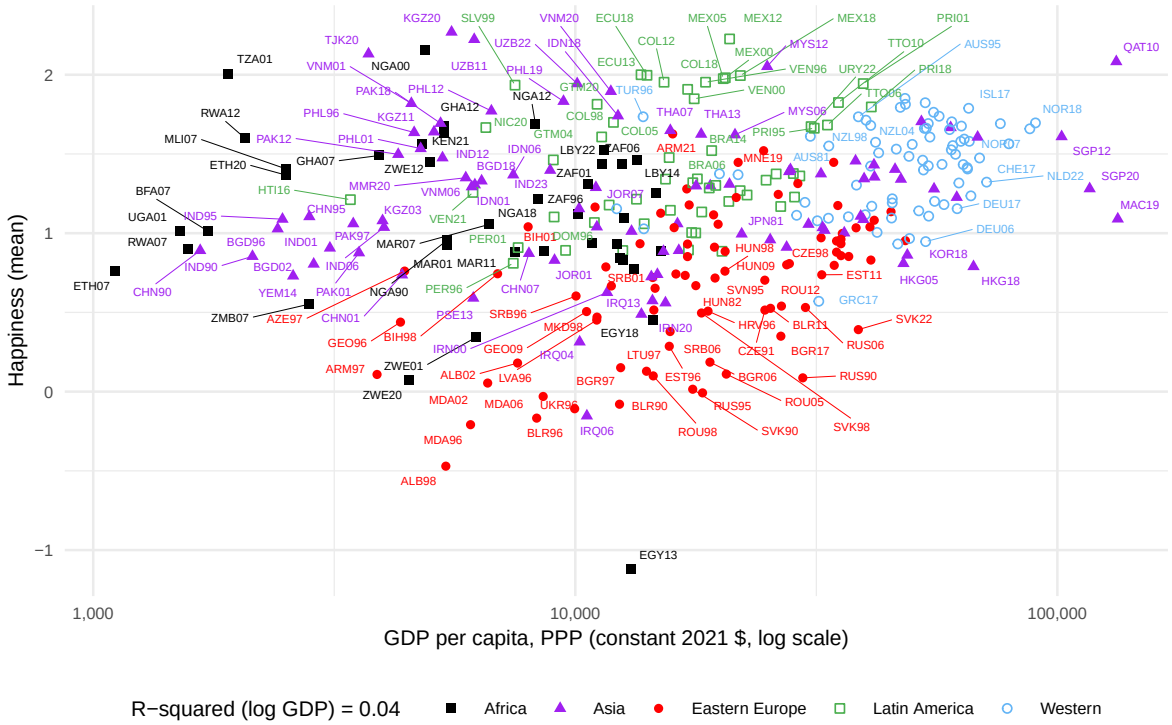


Figure 3: Mean happiness vs. GDP per capita (PPP), WVS/EVS 1981–2023.
Note: Answers coded +3 (*Very happy*), +1, −1, −3 (*Not at all happy*).

Well-being indicator	log GDP p.c.		Sextile	Income cluster				
	PPP	nominal	PPP	k=5 PPP	k=6 PPP	k=7 PPP	k=7 nominal	Region
Happy	0.14	0.20	0.20	0.19	0.20	0.20	0.23	0.30
Very Happy	0.00	0.00	0.03	0.03	0.04	0.05	0.04	0.33
Very Unhappy	0.07	0.10	0.12	0.10	0.13	0.13	0.14	0.14
V. Happy – V. Unhappy	0.00	0.01	0.04	0.04	0.04	0.05	0.05	0.33
Happiness (mean)	0.04	0.08	0.10	0.08	0.09	0.10	0.11	0.33
Satisfaction (mean)	0.23	0.29	0.23	0.21	0.22	0.23	0.29	0.41
Satisfied	0.32	0.38	0.32	0.30	0.32	0.32	0.38	0.39
Happy + Satisfied	0.27	0.35	0.30	0.28	0.30	0.30	0.35	0.38
Mean	0.14	0.18	0.17	0.15	0.17	0.17	0.20	0.33
Max	0.32	0.38	0.32	0.30	0.32	0.32	0.38	0.41

Table 1: Share of the variance of national well-being explained by income alone (R^2), WVS/EVS 1981–2023.
Note: R^2 of the regression of each well-being indicator (rows) on each income indicator (columns), over 342 WVS/EVS country-years (a few with missing GDP are excluded). The last column gives the R^2 of the regression on region dummies. The replication files include an Excel version of this table with the R^2 of all region classifications (`tables/main/r2_income_all_regions.xlsx`).

first split of a regression tree using log GDP and region as candidate variables is on region for all national well-being indicators.

Well-being indicator	log GDP p.c.		Sextile	Income cluster			
	PPP	nominal	PPP	k=5 PPP	k=6 PPP	k=7 PPP	k=7 nominal
Happy	0.29	0.36	0.38	0.35	0.37	0.38	0.41
Very Happy	0.00	0.01	0.07	0.06	0.08	0.10	0.10
Very Unhappy	0.28	0.38	0.43	0.40	0.47	0.47	0.50
V. Happy – V. Unhappy	0.01	0.02	0.09	0.07	0.09	0.11	0.11
Happiness (mean)	0.09	0.13	0.18	0.14	0.17	0.19	0.19
Satisfaction (mean)	0.32	0.38	0.32	0.29	0.31	0.32	0.38
Satisfied	0.43	0.49	0.43	0.41	0.43	0.44	0.49
Happy + Satisfied	0.39	0.46	0.42	0.39	0.41	0.42	0.47
Mean	0.23	0.28	0.29	0.26	0.29	0.30	0.33
Max	0.43	0.49	0.43	0.41	0.47	0.47	0.50

Table 2: Share of explained variance due to income rather than region (LMG), WVS/EVS 1981–2023.

Note: Share of the R^2 of Equation (3) attributed to income by the LMG (Shapley) decomposition, Equation (4). Values below 0.5 indicate that region is the better predictor.

The result is not an artifact of the larger number of parameters of the region model. Out of sample, region predicts better than income for every indicator (Online Appendix Table A3): for mean satisfaction, the leave-one-country-out R^2 is 0.37 with region, against 0.21 with log GDP and 0.23 with the best income measure (Cluster k=7 nominal), and for several happiness indicators, income has no out-of-sample predictive power at all (negative cross-validated R^2). Across the 952 estimates of the robustness checks (indicator $\times$ income measure $\times$ specification, where a specification is a row of Online Appendix Table A4: a sample such as a given wave, a weighting scheme, a GDP series, or a region classification), region is the better predictor in 88% of cases (79% with the best income measure). The result holds in each wave except wave 4 (1999–2004, 40 countries), where income and region perform similarly; when countries are weighted by population; without the EVS surveys; without the pandemic years; without imputed GDP values; and with the previous vintage of GDP data.

3.4 Where the region advantage comes from

Table 3 shows that the region advantage rests on two well-known "anomalies": Latin America, happier than its income would predict (Graham & Lora, 2009; Rojas, 2016), and Eastern Europe, unhappier (Guriev & Zhuravskaya, 2009). Region remains the better predictor when either one is excluded (in 70% of cases without Latin America and 73%

without Eastern Europe), but not when both are (27%). Consistently, the result holds with classifications that isolate both groups (exact UN groups and six regions: 100%), weakens with the World Bank regions, which isolate Latin America but merge Eastern and Western Europe (55%), and disappears with continents, which isolate neither (48%).

Region classification	Groups	Obs.	R^2 income (log PPP)	R^2 region	Share due to income (log PPP)	Region better (share of cases)
5 UN regional groups (main)	5	340	0.14	0.33	0.23	0.98
Exact UN regional groups	5	340	0.14	0.33	0.23	1.00
6 regions (Middle East separate)	6	340	0.14	0.32	0.22	1.00
World Bank regions	7	340	0.14	0.19	0.37	0.55
Continents	6	340	0.14	0.17	0.40	0.48
Main, w/o Latin America	4	285	0.15	0.30	0.26	0.70
Main, w/o Eastern Europe	4	256	0.16	0.22	0.29	0.73
Main, w/o both	3	201	0.18	0.20	0.35	0.27

Table 3: Income vs. region by region classification and without Latin America or Eastern Europe (WVS/EVS).

Note: Averages over the eight well-being indicators. R^2 and share of explained variance due to income (Equation (4)) with log GDP p.c. (PPP). "Region better": share of the indicator $\times$ income-measure combinations for which $s < 0.5$. Region classifications are described in Section 2.4.

3.5 Income predicts national well-being as well as region in Gallup

The conclusion does not carry over to Gallup data (Figure 4 and Table 4). In Gallup's latest observation for each of 168 countries, log GDP explains 61% of the variance of the mean ladder (95% CI [52%; 69%]), against 48% for region ([38%; 61%]), and income accounts for 60% of the explained variance ([49%; 68%]). Income is the better predictor for the mean and for the shares of respondents in the middle and bottom of the distribution, with all income measures. Over all Gallup specifications, region is the better predictor in only 34% of cases, mostly for indicators of the upper tail of the distribution (shares answering 9–10 or 10), which are almost unrelated to income—echoing the weak relation between income and the share of *Very happy* people in the WVS. With the latest WHR data (2023–2025), log GDP explains 63% of the variance of the ladder against 55% for region. Whether "GDP is a poor predictor of national well-being" therefore depends on which of the two datasets one trusts, a question I turn to in Section 4.

3.6 Other correlates and within-country evidence

What do regions capture? The average perceived freedom of choice and control over one's life alone explains 63% of the variance of mean satisfaction; in a Shapley decomposition

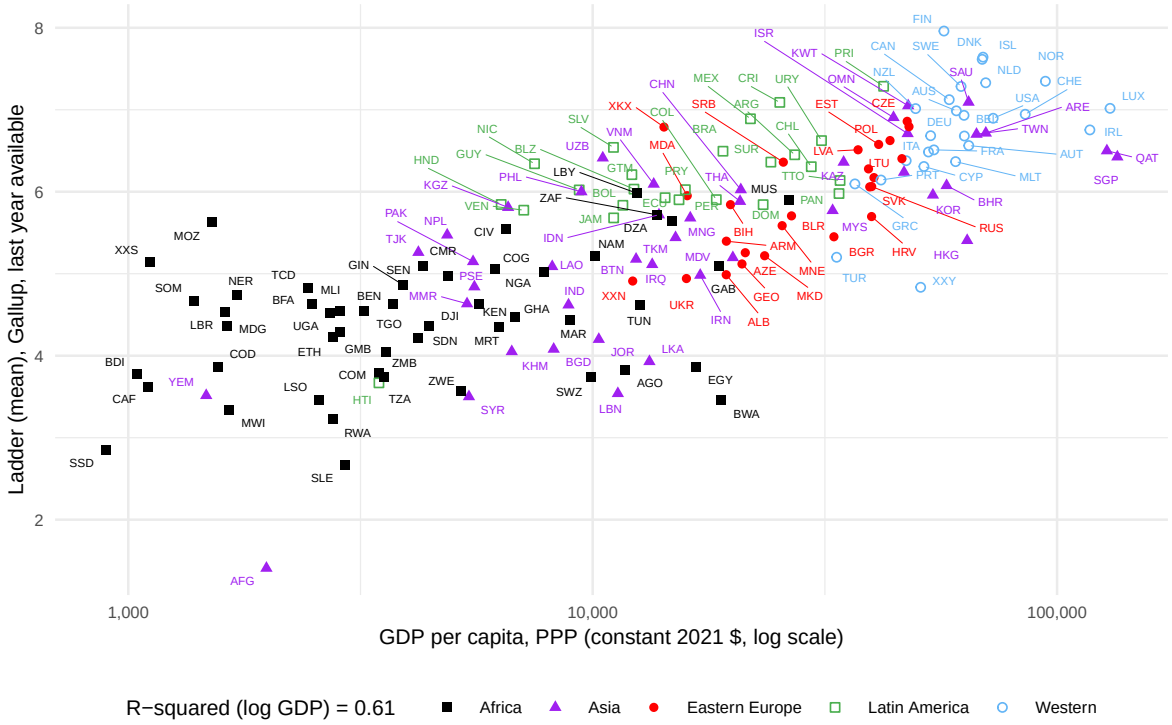


Figure 4: Mean ladder vs. GDP per capita (PPP), Gallup, last observation per country.

Well-being indicator	log GDP p.c.		Sextile	Income cluster			
	PPP	nominal	PPP	k=5 PPP	k=6 PPP	k=7 PPP	k=7 nominal
Satisfied	0.58	0.61	0.59	0.58	0.60	0.59	0.60
Very satisfied (8–10)	0.44	0.51	0.49	0.47	0.49	0.51	0.53
Extremely satisfied (9–10)	0.16	0.25	0.24	0.27	0.22	0.30	0.28
Completely satisfied (10)	0.02	0.02	0.08	0.09	0.06	0.08	0.08
Unsatisfied (0/1–4)	0.61	0.60	0.60	0.59	0.61	0.60	0.59
Dissatisfied (0/1–2)	0.66	0.63	0.67	0.66	0.67	0.67	0.65
Satisfaction (mean)	0.60	0.62	0.59	0.58	0.60	0.59	0.60
Mean	0.44	0.46	0.47	0.46	0.47	0.48	0.48
Max	0.66	0.63	0.67	0.66	0.67	0.67	0.65

Table 4: Share of explained variance due to income rather than region (LMG), Gallup, last observation per country.

Note: 168 countries, last Gallup observation available (2006–2023). Same definitions as Table 2; values above 0.5 indicate that income is the better predictor.

with income and region, it accounts for 42 percentage points, against 21 for region and 11 for income (Online Appendix Table A5). This variable is itself a subjective evaluation, so the correlation should not be read causally, but it suggests that non-material dimensions of life are central to regional differences (Inglehart et al., 2008). Tolerance of homosexuality (24%), the importance of God (1%) and GDP growth over the previous five years (2%) explain much less. Other measures of material living standards do not predict better than GDP: median income (World Bank, 2026a) explains 19% of the variance of mean satisfaction (against 23% for log GDP in the same country-years), and household final consumption expenditure per capita 30%. In Shapley decompositions that include them along with GDP and region, region remains the largest contributor.

The cross-sectional relation should not be confused with the relation over time. Within countries (country fixed effects, 312 observations), mean satisfaction increases with log GDP per capita, with a slope of 2.31 per unit of $\log_{10}$ GDP, larger than the cross-sectional slope (1.14), although income growth explains only 0.34 of the within-country variance (Online Appendix Table A6). These associations are consistent with Stevenson & Wolfers (2008) and Sacks et al. (2012) rather than with a strict version of the Easterlin paradox (Easterlin, 1974; Easterlin et al., 2010), although they may reflect common shocks (e.g. transition recessions) rather than the effect of income per se. Across countries, Deaton (2008) found that, controlling for the level of GDP per capita, growth was negatively associated with life satisfaction in the 2006 Gallup World Poll. His specification replicates with the present data (2006 ladder on log GDP in 2003 and average annual growth in 2000–2003: coefficient -5.50, $p = .017$). But, as Deaton noted, this result is driven by former communist countries, which combined fast growth with very low satisfaction (without them, the coefficient is -0.53 and not significant, $p = .855$). It does not extend to other periods either: over all country-years, controlling for log GDP, growth over the previous five years is not significantly related to the Gallup ladder ($p = .957$) and is slightly positively related to WVS satisfaction (0.039 points per percentage point, $p = .044$).

3.7 What are the happiest countries?

Rankings of countries depend on the indicator. Looking at all waves combined, the happiest country-year is Kyrgyzstan 2020 (3 indicators, including mean happiness), and Finland 1981, Malaysia 2012, Mexico 2012, Qatar 2010 and Vietnam 2020 for one indicator each. Counting the happiest country-year for each indicator and each wave (and all waves combined), Switzerland appears most often (10 times out of 64), followed by Mexico (9) and Kyrgyzstan (6) (Online Appendix Table A7). The happiest country-years are

Western (25), Latin American (19), Asian (15) or African (5). Similarly, with Gallup data on emotions, Blanchflower & Bryson (2024) find that the top-ranked country or U.S. state differs across measures: Denmark for the ladder, Hawaii for enjoyment, Paraguay for smiling, Bhutan for being well-rested, and Vietnam, Taiwan, Uzbekistan and Finland for the lowest prevalence of pain, sadness, worry and anger, respectively. Beyond nationally representative surveys, high life satisfaction is also reported in small-scale societies with very low monetary incomes (Galbraith et al., 2024), and in the Global Flourishing Study, a broader index of flourishing is highest in Indonesia, Mexico and the Philippines and lowest in Japan (VanderWeele et al., 2025). The link between income and well-being seems to be that rich countries rarely experience low well-being, rather than that they reach the highest levels.

4 Why do Gallup and the WVS disagree? Wording vs. samples

4.1 The discrepancy

Restricting both datasets to the 187 country-years (99 countries) observed by both surveys the same year, log GDP explains 0.53 of the variance of Gallup's mean ladder but only 0.16 of WVS mean satisfaction, and income accounts for 51% of the variance explained jointly with region in Gallup against 24% in the WVS. The discrepancy found in Section 3 is thus not due to different country coverage. The gap between the two datasets decreases strongly with GDP (Figure 5): the slope is -1.08 per unit of $\log_{10}$ GDP (s.e. 0.14, clustered by country), with WVS answers exceeding Gallup answers by around two points in low-income countries, against less than half a point in high-income countries.

The two datasets differ in their question (ladder 0–10 vs. satisfaction 1–10) and in their samples in the broad sense defined in Section 2.5. The experiment allows me to measure how much of the discrepancy the question accounts for: first for the relation between income and well-being, which is the object of this paper (Section 4.2), then for levels (Section 4.3).

4.2 Does wording explain the discrepancy? Evidence from ten countries

The ten countries of the experiment provide a direct test. For each question variant, I compute the mean answer in each country and regress it on log GDP per capita across the ten

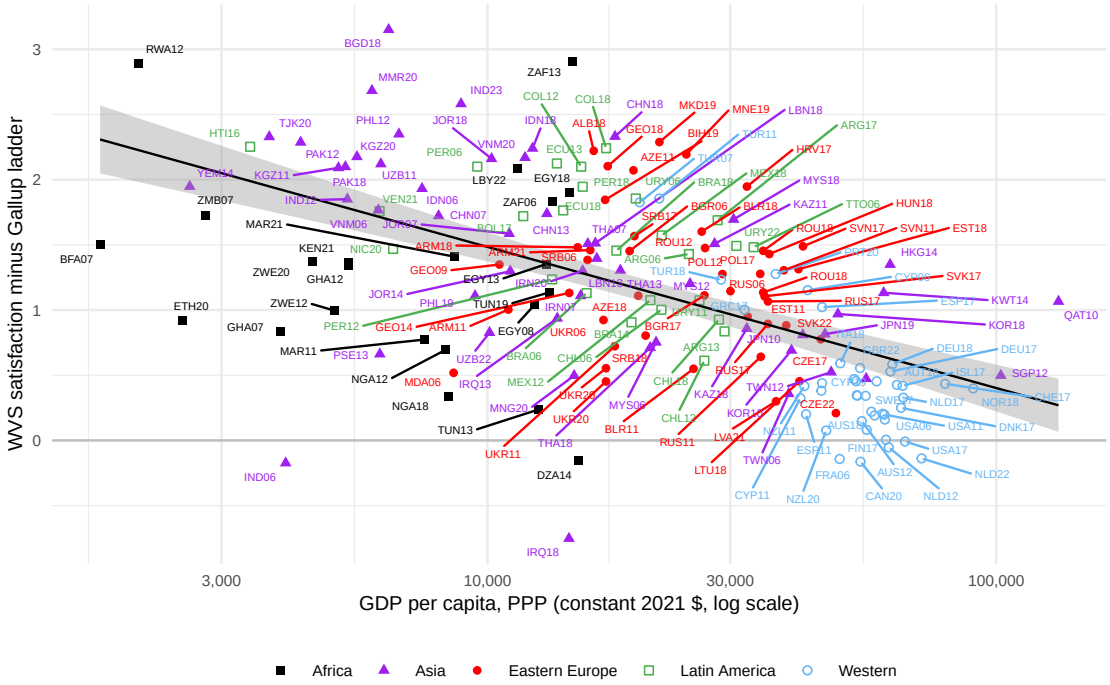


Figure 5: WVS satisfaction minus Gallup ladder in the same country-year, by GDP per capita.
Note: Mean answers on native scales (WVS: 1–10; Gallup: 0–10), 187 country-years observed by both surveys. Line: OLS fit with 95% confidence band.

countries, as for the Gallup and WVS country means (Table 5 and Figure 6). Each country thus has the same weight, whatever its sample size. For Gallup and the WVS, I use the latest WVS or EVS survey of each country (2017–2022, except Saudi Arabia, 2003) and the Gallup data of the same year (or the closest year, at most three years apart). Confidence intervals come from a bootstrap that resamples respondents in the three surveys.

I first compare R^2, which directly measure the predictive capacity of GDP per capita. In the past data, the ten countries reproduce the global discrepancy: log GDP explains 0.75 (95% CI [0.68; 0.80]) of the cross-country variance of Gallup's ladder but only 0.16 ([0.10; 0.24]) of WVS satisfaction. Asked to the same online sample, the two original questions reproduce much of this contrast: log GDP explains 0.76 ([0.52; 0.83]) of the variance of the ladder 0–10 but 0.31 ([0.16; 0.45]) of satisfaction 1–10. Pooling both scales of each wording, the R^2 is 0.61 for the ladder and 0.28 for satisfaction. The wording thus accounts for 56% (95% CI [22%; 82%]) of the Gallup–WVS difference in R^2 (76% with the two original variants only, [31%; 107%]), and samples for the rest. In other words, the new data lie midway between Gallup and the WVS. Pooling the four variants, log GDP explains 0.43 of the variance, and income accounts for 0.58 of the variance explained jointly with region, halfway between Gallup (0.78) and the WVS (0.35).

	Past data		Fabre (2025)						
Data	Gallup	WVS/EVS							
Question	Ladder 0–10	Satisfaction 1–10	Ladder 0–10	Ladder 1–10	Satisfaction 0–10	Satisfaction 1–10	Ladder (both scales)	Satisfaction (both scales)	All four variants
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\log_{10}$ GDP p.c. (PPP)	4.20***	1.36	4.35**	2.57	3.21	4.05	3.60*	3.86	3.73*
	(0.80)	(1.44)	(1.05)	(1.36)	(2.03)	(2.09)	(1.23)	(2.14)	(1.55)
Observations (countries)	10	10	10	10	10	10	10	10	10
R^2	0.75	0.16	0.76	0.33	0.23	0.31	0.61	0.28	0.43
95% CI of R^2	[0.68; 0.80]	[0.10; 0.24]	[0.52; 0.83]			[0.16; 0.45]	[0.44; 0.70]	[0.16; 0.37]	[0.32; 0.51]
R^2 region	0.29	0.34	0.14	0.50	0.36	0.29	0.33	0.32	0.34
Share due to income	0.78	0.35	0.91	0.36	0.35	0.52	0.71	0.45	0.58

Table 5: Relation between GDP per capita and national well-being in the ten countries of the experiment: past vs. new data.

Note: Each column is an OLS regression of the country mean answer on $\log_{10}$ GDP per capita (PPP) across the 10 countries of the experiment, one observation per country. Columns (1)–(2): latest WVS/EVS survey and Gallup the same year, GDP of that year. Columns (3)–(9): Fabre (2025), mean answer by variant (native scale), GDP of 2025; "both scales" and "all four variants": average of the variant means, 1–10 answers being linearly stretched to 0–10. Heteroskedasticity-robust (HC1) standard errors in parentheses. * $p < .05$, ** $p < .01$, *** $p < .001$. 95% CI of R^2: bootstrap resampling respondents (not computed for the two crossed variants). R^2 region and share due to income: see Section 3.1 (regions: Western, Asia, Eastern Europe); a share above 0.5 indicates that income is the better predictor.

Slopes tell a different story. For both questions, the estimated income gradient is close to Gallup's (4.20): 4.35 with the ladder and 4.05 with the satisfaction question (3.60 and 3.86 when pooling scales), whereas the WVS estimate is three times smaller (1.36). With ten countries, these slopes are imprecisely estimated: those of the satisfaction question and of the WVS are not significantly different from zero ($p = .089$ and $p = .373$, respectively). The part of the Gallup–WVS difference in slopes that is not due to the question is 2.54 (95% CI [0.90; 4.18]). How can the satisfaction question yield a lower R^2 but the same slope as the ladder? With a single regressor, $R^2 = \beta^2 \mathrm{var}(x)/\mathrm{var}(y)$: for a given slope, the R^2 is lower when country means are more dispersed around the income gradient. The satisfaction question does not appear to weaken the gradient, but it produces larger country-specific deviations from it—most visibly in Japan, whose mean satisfaction (5.00 on the 1–10 scale) is 1.4 points below that of any other country despite its high income, whereas its mean ladder (5.33) is only 0.2 points below the next lowest (Online Appendix Table A9). These country-specific deviations are present both in the WVS and in the new survey data. By contrast, the flatter WVS gradient appears to be specific to its samples.

However, predictive capacity is measured by the R^2 rather than by the slope. Taken together, the results suggest that the wording accounts for about half of the difference in predictive capacity between Gallup and the WVS, through larger country-specific deviations in answers to the satisfaction question, while differences between samples account for the other half and for most of the difference in the income gradient. Two caveats apply. First, with only ten countries, R^2 and slopes are imprecise. The bootstrap confidence intervals reflect the sampling of respondents and treat the ten countries as given (Table 5

reports the standard errors of slopes across countries). The small number of observations also makes results sensitive to individual countries: for instance, excluding Japan raises the R^2 of Gallup to 0.91 and lowers the share of the Gallup–WVS difference in R^2 due to wording to 38%. Second, the new data say little about income versus region, since seven of the ten countries are Western.

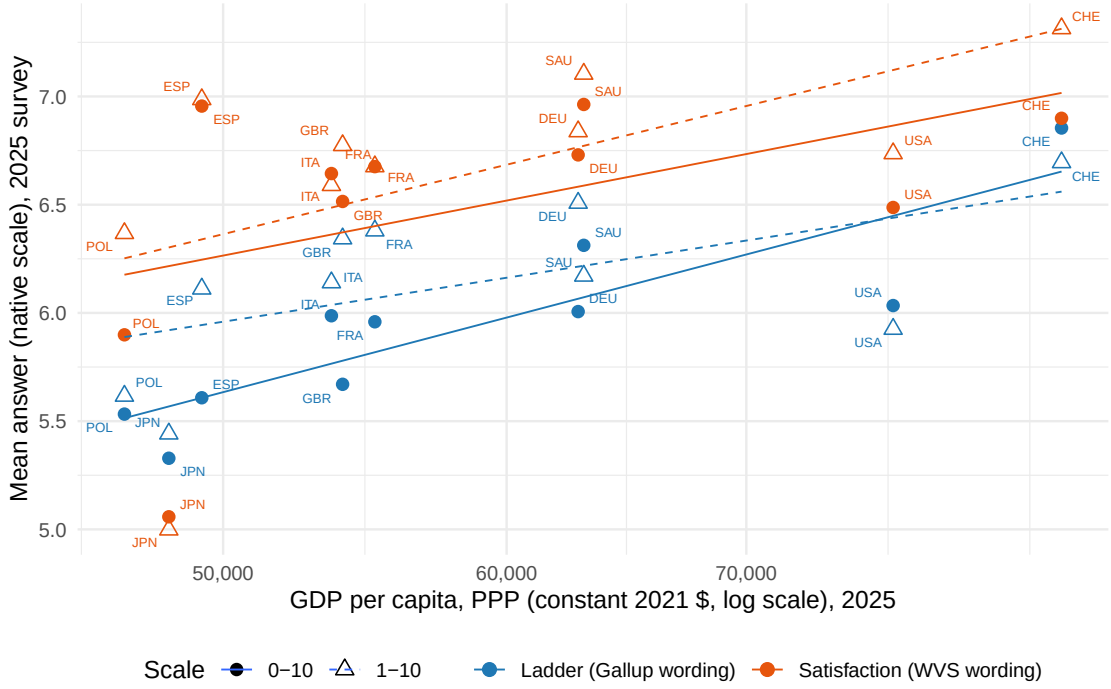


Figure 6: Mean life evaluation vs. GDP per capita (PPP), by question variant, Fabre (2025).
Note: Mean answer on the native scale of each variant (0–10 or 1–10), ten countries surveyed in 2025. Colors: wording; shapes and line types: scale. Lines: OLS fits by variant.

4.3 Wording and the levels of well-being

Although levels matter less than correlations for our question, the experiment also explains why the WVS yields higher levels than Gallup. Online Appendix Table A8 reports regressions of the answer on the randomized treatments, with country fixed effects, survey weights and heteroskedasticity-robust standard errors. The ladder wording shifts answers by -0.45 points on average (s.e. 0.05): respondents rate their lives lower on the "best possible life" ladder than when asked how satisfied they are, in line with evidence that the ladder evokes power and wealth more than life satisfaction does (Nilsson et al., 2024).

The wording effect varies markedly across countries (Figure 7; $\chi^2(9) = 93.4$, $p < .001$). Japan is the only country where the ladder yields *higher* answers than the satisfaction

question (by 0.36 points); elsewhere, the wording effect ranges from -1.12 (Spain) to -0.34 points. Besides, I find no evidence that the wording changes the income gradient *within* countries: the interaction between the ladder wording and the respondent's income quartile is small and not significant (-0.004, 95% CI [-0.086; 0.077], $p = .919$), compared with a gradient of 0.69 points per quartile.

Answers on the 1–10 scale are 0.14 points higher than on the 0–10 scale (s.e. 0.05). Once 1–10 answers are linearly stretched to 0–10 with $x \mapsto (x-1) \times 10/9$, which maps 1 to 0 and 10 to 10, the scale effect becomes -0.26: the stretch lowers each answer x by $(10-x)/9$, i.e. by about 0.4 points at the average answer. The interaction between wording and scale is not significant (0.008, s.e. 0.092, $p = .930$).

To relate these effects to the Gallup–WVS gap, I define for each country c the observed gap $D_c = W_c - G_c$, where W_c is mean satisfaction in the WVS and G_c the mean ladder in Gallup the same year (as in Section 4.2), and the question effect $Q_c = F_c(\text{satisfaction 1–10}) - F_c(\text{ladder 0–10})$, the difference between the mean answers to the two original variants in the experiment, measured on the same sample. The residual $R_c = D_c - Q_c$ is the part of the gap due to samples in the broad sense. On average, WVS answers exceed Gallup answers by $D = 0.60$ points, and the question effect is even larger: $Q = 0.71$ points (Online Appendix Tables A9 and A10). The question thus fully accounts for the higher level of WVS answers in high-income countries, and the residual is small ($R = -0.11$, 95% CI [-0.26; 0.04]). Since the two components move in opposite directions, another way to summarize the level difference is to compare the size of the two movements: the question accounts for $|\bar{Q}|/(|\bar{Q}| + |\bar{R}|) = 86\%$ (95% CI [76%; 99%]) of the gross difference in levels.

The heterogeneous question effect does not, however, explain why the gap varies across countries. Since $D_c = Q_c + R_c$, the cross-country variance of the gap splits into

$$\text{var}(D) = \text{cov}(D, Q) + \text{cov}(D, R).$$

The share attributable to the question, $\text{cov}(D, Q)/\text{var}(D)$, is only 11% (95% CI [$-33\%$; 55%]): countries with a larger question effect do not tend to have a larger gap. Accordingly, predicting each country's Gallup mean by subtracting its own question effect Q_c from its WVS mean does worse (root mean squared error: 0.54) than simply subtracting the average gap (0.35).

Samples asked the same question also differ markedly. Gallup respondents (2023–2025) rate their lives 0.75 points higher on the ladder 0–10 than online respondents in 2025, and WVS respondents 0.75 points higher on satisfaction 1–10 (even in the United States,

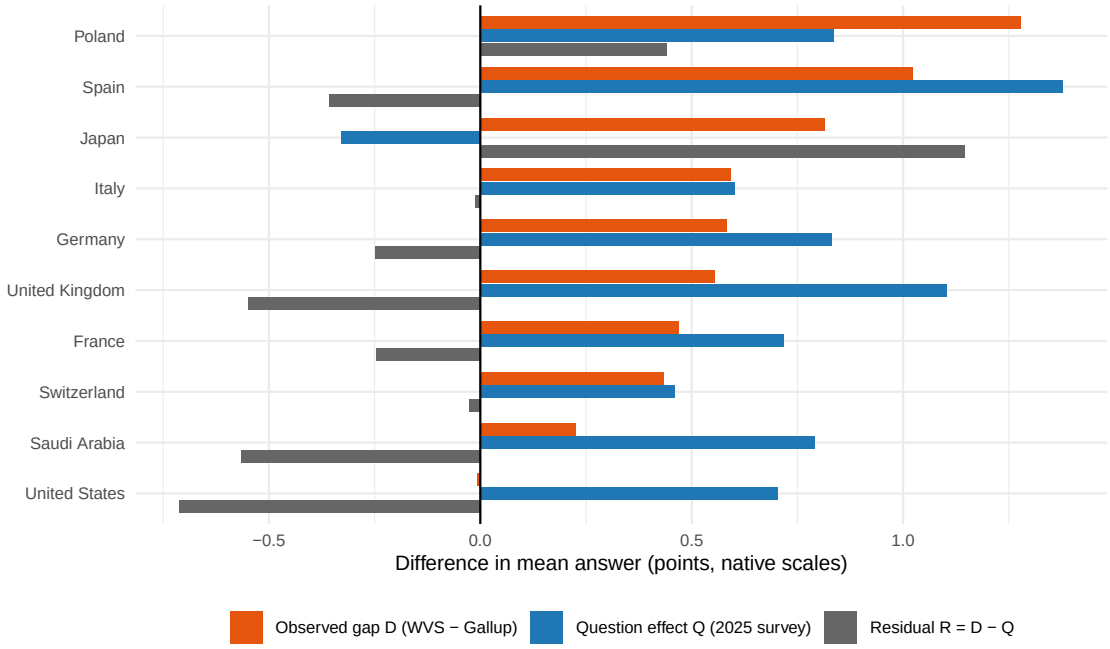


Figure 7: Observed WVS–Gallup gap, question effect and residual by country.
Note: D: mean satisfaction (WVS/EVS, 1–10) minus mean ladder (Gallup, 0–10), same year. *Q*: mean answer to satisfaction 1–10 minus mean answer to ladder 0–10, Fabre (2025). $R = D - Q$.

where the 2017 WVS was itself conducted online: 0.49 points). And in the 10 countries surveyed separately by the EVS and the WVS in 2017–2022, with the same question, mean satisfaction differs by 0.28 points on average (up to 0.53).

In sum, the question explains why WVS answers are higher than Gallup answers on average, but not why this gap varies across countries. Both the levels and their variation across countries also depend heavily on the samples.

4.4 Which dataset has more representative samples?

If samples matter, which dataset is closer to the truth? There is no clear-cut argument in favor of either dataset, nor of the online sample used in the experiment. Gallup applies a centralized, standardized design: probability samples of about 1,000 adults per year, drawn by random-digit dialing where telephone coverage is high and by multistage area sampling elsewhere, with random selection of the respondent within the household and post-stratification weights on gender, age and, where possible, education (Gallup, 2023). But its samples are small; its face-to-face samples select households with replacement of non-responding households; response rates are not published; interviews switched from face-to-face to telephone during the pandemic; and in 2023, a fifth of the interviews in 27 high-income countries came from web panels, most of them opt-in (Gallup, 2023).

Its subgroup estimates have also been questioned (Blanchflower, 2024). Besides, Gallup microdata are not public, so that the representativeness of its samples cannot be assessed independently; lacking quotas, its samples could under-represent hard-to-reach groups such as less educated or rural people. The WVS requires probability samples of at least 1,200 respondents but allows quota selection at the last stage, and its sampling designs, modes and survey years vary across countries (World Values Survey Association, 2023).

I test the representativeness of WVS/EVS samples directly (Online Appendix Table A11). Compared with population benchmarks (Barro & Lee, 2013; World Bank, 2026b), WVS samples over-represent tertiary-educated adults (aged 25 and over) by a median factor of 1.98 in countries with a GDP per capita below 10,000 dollars, against 1.05 in countries above 30,000 dollars (295 surveys). They also over-represent urban residents in poorer countries (median ratio 2.19 vs. 1.22) and under-represent people aged 65 or more (0.64 vs. 1.01). The internal criterion of Kohler (2007)—women should make up half of married or cohabiting respondents—is violated at the 5% level in 36% of surveys. WVS samples are thus less representative in poorer countries, as noted by Deaton (2008), which is where their gap with Gallup is largest. However, correcting these observable imbalances changes little. The over-representation of the tertiary-educated is not significantly related to the gap: in a regression of the WVS–Gallup gap on log GDP and the log of the ratio between the sample and population shares of tertiary-educated adults, the coefficient on this log ratio is -0.07 (s.e. 0.07, $p = .311$). And reweighting each WVS sample to the benchmark share of tertiary-educated adults barely changes its relation with income: in the country-years also covered by Gallup, the R^2 of mean satisfaction on log GDP goes from 0.19 to 0.20, against 0.54 for Gallup. The discrepancy thus stems from unobserved differences in who responds, how, and in which context, which the available evidence does not allow me to attribute to one dataset rather than the other.

5 Discussion

5.1 Income vs. region

Region is not a mechanism: it bundles culture, history, institutions, social ties, and possibly cross-cultural differences in how people use response scales (Chen et al., 1995; Diener et al., 2003; Exton et al., 2015). If regional differences partly reflect response styles, region predicts *measured* rather than *experienced* well-being. Studies of migrants suggest that national differences partly reflect a cultural determinant of well-being rather than trans-

lation artifacts (Exton et al., 2015; Senik, 2014), and cultural values such as generalized trust and morality are associated with higher happiness among Europeans (Conzo et al., 2017), and the low well-being of transition countries has been related to the deterioration of institutions, economic conditions, and public goods provision (Guriev & Zhuravskaya, 2009). The present analysis cannot separate these (or other) interpretations. Anchoring vignettes—short descriptions of hypothetical people whose lives respondents rate on the same scale as their own, which reveal how different groups use the scale (Kapteyn et al., 2010)—or measures of emotional well-being would help. Moreover, the advantage of region rests on Latin America and Eastern Europe. The fairest summary is that GDP per capita leaves much of the cross-country variation unexplained, and that a few large regional deviations explain more of it than income does in the WVS.

The modest cross-country role of income echoes within-country evidence. Studies comparing people within countries find a small effect of income on life satisfaction relative to other circumstances such as employment or marital status (Blanchflower & Oswald, 2004; Helliwell, 2003); Kahneman et al. (2006) argue that even these estimates overstate the role of income, because of a "focusing illusion" when people evaluate their lives; and in German panel data, about two thirds of the effect of an income increase on happiness vanish within four years (Di Tella et al., 2010). Conversely, emotional well-being increases with log income for most people in the United States (Killingsworth et al., 2023), and lower-income people report more frequent boredom in 30 countries (Pirla et al., 2025), and the within-country association between GDP and satisfaction documented above shows that the weak cross-sectional relation does not imply that income is irrelevant.

5.2 Wording vs. samples

The question matters for levels and for predictive capacity: the satisfaction question yields higher answers and larger country-specific deviations from the income gradient than the ladder. The question specification reproduces half of the Gallup–WVS difference in R^2. But it does not explain the flatter income gradient of the WVS, nor the cross-country pattern of the gap in levels, which reflect differences between samples in the broad sense: sampling frames, interview modes (Conti & Pudney, 2011; Dolan & Kavetsos, 2016), questionnaire context (Deaton, 2012; Deaton & Stone, 2016), translations, and time.

Several limitations should be kept in mind. First, the experiment covers only ten high-income countries, whereas the gap varies most across income levels. The experiment cannot rule out that the wording effect is different in low-income countries—for instance it could be larger if the "best possible life" evokes material comparisons more strongly

where material deprivation is widespread. Only an experiment in low- and middle-income countries can settle this. Second, the residual lumps together several factors that the design cannot separate; in particular, the question effect was measured online, toward the end of the questionnaire, and may differ from the effect that would be obtained at the beginning of an interview conducted face-to-face or by phone. Third, the comparison with past data assumes that the question effect measured in 2025 applies to the surveys of 2017–2022 (2003 for Saudi Arabia).

6 Conclusion

GDP per capita is a weaker predictor of national well-being than is often assumed, but how weak depends on the data. In the WVS, it explains a small share of the cross-country variance of life satisfaction, less than a country's world region, and an even smaller share of the variance of self-declared happiness, which region predicts nearly as well as life satisfaction. The richest countries are not necessarily the happiest: the country-year with the highest happiness mean score in the WVS is Kyrgyzstan in 2020, with a GDP per capita of about 5,500 dollars; and many middle-income countries, especially in Latin America, are as satisfied as the richest ones. In the Gallup World Poll, however, income predicts life evaluations as well as region does. A survey experiment in ten high-income countries shows that both the question and the samples contribute to this discrepancy: asked to the same sample, the ladder used by Gallup is better predicted by GDP than the WVS satisfaction question, reproducing about half of the Gallup–WVS difference in predictive capacity, but the estimated income gradient is close to Gallup's for both questions, whereas the WVS gradient is much flatter.

These findings have implications for research and policy. For research, cross-country conclusions about the income–well-being relationship should be checked across datasets and questions, as no dataset is clearly more representative. The gap between datasets is largest in low-income countries, where replicating the experiment face-to-face is the natural next step. Other research avenues include testing whether region also predicts emotions better than income (Blanchflower & Bryson, 2024; Killingsworth et al., 2023), and, despite evidence against translation issues (Diener & Suh, 2000), using anchoring vignettes to rule out response styles as an explanation of regional differences. For policy, the results suggest that absolute income is less determining for national well-being than is often thought: rich countries rarely have low satisfaction but sometimes have lower happiness than average, while middle-income countries can be as happy as the richest, so that growth is neither necessary nor sufficient for high well-being. Well-being itself,

rather than growth, an imprecise proxy, should be the primary target when evaluating policies. And since non-material dimensions of life seem central to the regional differences in well-being, understanding their mechanisms deserves more attention (Frijters et al., 2020; Stiglitz et al., 2009).

References

R. J. Barro & J. W. Lee. A new data set of educational attainment in the world, 1950–2010. *Journal of Development Economics*, 2013. Link. 22

M. Berlin & F. Fors Connolly. The association between life satisfaction and affective well-being. *Journal of Economic Psychology*, 2019. Link. 3

C. Bjørnskov. How comparable are the Gallup World Poll life satisfaction data? *Journal of Happiness Studies*, 2010. Link. 3

D. G. Blanchflower. Is the Gallup World Poll reliable? After Babel, 2024. Link. 22

D. G. Blanchflower & A. Bryson. Wellbeing rankings. *Social Indicators Research*, 2024. Link. 16, 24

D. G. Blanchflower & A. J. Oswald. Well-being over time in Britain and the USA. *Journal of Public Economics*, 2004. Link. 23

T. N. Bond & K. Lang. The sad truth about happiness scales. *Journal of Political Economy*, 2019. Link. 3, 5

H. Cantril. *The Pattern of Human Concerns*. Rutgers University Press, New Brunswick, NJ, 1965. 5

C. Chen, S.-y. Lee, & H. W. Stevenson. Response style and cross-cultural comparisons of rating scales among East Asian and North American students. *Psychological Science*, 1995. Link. 22

G. Conti & S. Pudney. Survey design and the analysis of satisfaction. *Review of Economics and Statistics*, 2011. Link. 23

P. Conzo, A. Aassve, G. Fuochi, & L. Mencarini. The cultural foundations of happiness. *Journal of Economic Psychology*, 2017. Link. 3, 23

A. Deaton. Income, health, and well-being around the world: Evidence from the Gallup World Poll. *Journal of Economic Perspectives*, 2008. Link. 2, 3, 10, 15, 22

A. Deaton. The financial crisis and the well-being of Americans. *Oxford Economic Papers*, 2012. Link. 23

A. Deaton & A. A. Stone. Understanding context effects for a measure of life evaluation: How responses matter. *Oxford Economic Papers*, 2016. Link. 3, 23

R. Di Tella, J. Haisken-De New, & R. MacCulloch. Happiness adaptation to income and to status in an individual panel. *Journal of Economic Behavior & Organization*, 2010. Link. 23

E. Diener & E. M. Suh, editors. *Culture and Subjective Well-Being*. MIT Press, Cambridge, MA, 2000. Link. 3, 24

E. Diener, S. Oishi, & R. E. Lucas. Personality, culture, and subjective well-being: Emotional and cognitive evaluations of life. *Annual Review of Psychology*, 2003. Link. 22

P. Dolan & G. Kavetsos. Happy talk: Mode of administration effects on subjective well-being. *Journal of Happiness Studies*, 2016. Link. 3, 23

R. A. Easterlin. Does economic growth improve the human lot? some empirical evidence. In P. A. David & M. W. Reder, editors, *Nations and Households in Economic Growth*. Academic Press, New York, 1974. Link. 3, 15

R. A. Easterlin, L. A. McVey, M. Switek, O. Sawangfa, & J. S. Zweig. The happiness–income paradox revisited. *Proceedings of the National Academy of Sciences*, 2010. Link. 3, 15

EVS/WVS. Joint EVS/WVS 2017–2022 dataset (joint EVS/WVS). GESIS, Cologne, ZA7505, data file version 5.0.0 (2024-06-24), 2024. Link. 4

C. Exton, C. Smith, & D. Vandendriessche. Comparing happiness across the world: Does culture matter? OECD Statistics Working Papers 2015/04, OECD Publishing, 2015. Link. 3, 22, 23

A. Fabre. Public acceptance of international redistribution in high-income countries. 2025. Link. 7, 8

P. Frijters, A. E. Clark, C. Krekel, & R. Layard. A happy choice: Wellbeing as the goal of government. *Behavioural Public Policy*, 2020. Link. 25

E. D. Galbraith, C. Barrington-Leigh, S. Miñarro, S. Álvarez-Fernández, E. M. N. A. N. Attoh, P. Benyei, L. Calvet-Mir, R. Carmona, R. Chakauya, Z. Chen, F. Chengula, Á. Fernández-Llamazares, D. García-del Amo, M. Glauser, T. Huanca, A. E. Izquierdo, A. B. Junqueira, M. Lanker, X. Li, J. Mariel, M. D. Miara, V. Porcher, A. Porcuna-Ferrer, A. Schlingmann, R. Seidler, U. B. Shrestha, P. Singh, M. Torrents-Ticó, T. Ulambayar, R. Wu, & V. Reyes-García. High life satisfaction reported among small-scale societies with low incomes. *Proceedings of the National Academy of Sciences*, 2024. Link. 16

Gallup. Gallup global research methodology (world poll), 2022. Link. 5

Gallup. Worldwide research methodology and codebook (updated october 2023). Gallup, Washington, DC, 2023. Link. 21

C. Graham & E. Lora, editors. *Paradox and Perception: Measuring Quality of Life in Latin America*. Brookings Institution Press, Washington, DC, 2009. 3, 12

U. Grömping. Estimators of relative importance in linear regression based on variance decomposition. *The American Statistician*, 2007. Link. 2, 9

S. Guriev & E. Zhuravskaya. (un)happiness in transition. *Journal of Economic Perspectives*, 2009. Link. 3, 12, 23

C. Haerpfer, R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, & B. Puranen. World values survey trend file (1981–2022) cross-national data-set. JD Systems Institute & WVSA Secretariat, Madrid & Vienna. Data file version 3.0.0, 2022. Link. 4

J. F. Helliwell. How's life? combining individual and national variables to explain subjective well-being. *Economic Modelling*, 2003. Link. 23

R. Inglehart & H.-D. Klingemann. Genes, culture, democracy, and happiness. In E. Diener & E. M. Suh, editors, *Culture and Subjective Well-Being*. MIT Press, Cambridge, MA, 2000. Link. 2, 5

R. Inglehart, R. Foa, C. Peterson, & C. Welzel. Development, freedom, and rising happiness: A global perspective (1981–2007). *Perspectives on Psychological Science*, 2008. Link. 2, 10, 15

International Monetary Fund. World economic outlook database (DataMapper). Indicators PPPPC, NGDPDPC and LP, accessed 27 September 2026, 2026. Link. 6

A. T. Jebb, L. Tay, E. Diener, & S. Oishi. Happiness, income satiation and turning points around the world. *Nature Human Behaviour*, 2018. Link. 3

D. Kahneman & A. Deaton. High income improves evaluation of life but not emotional well-being. *Proceedings of the National Academy of Sciences*, 2010. Link. 2

D. Kahneman & A. B. Krueger. Developments in the measurement of subjective well-being. *Journal of Economic Perspectives*, 2006. Link. 3

D. Kahneman, A. B. Krueger, D. Schkade, N. Schwarz, & A. A. Stone. Would you be happier if you were richer? A focusing illusion. *Science*, 2006. Link. 23

C. Kaiser & A. J. Oswald. The scientific value of numerical measures of human feelings. *Proceedings of the National Academy of Sciences*, 2022. Link. 3

A. Kapteyn, J. P. Smith, & A. van Soest. Life satisfaction. In E. Diener, J. F. Helliwell, & D. Kahneman, editors, *International Differences in Well-Being*. Oxford University Press, Oxford, 2010. Link. 23

M. A. Killingsworth, D. Kahneman, & B. Mellers. Income and emotional well-being: A conflict resolved. *Proceedings of the National Academy of Sciences*, 2023. Link. 23, 24

U. Kohler. Surveys from inside: An assessment of unit nonresponse bias with internal criteria. *Survey Research Methods*, 2007. Link. 22

R. H. Lindeman, P. F. Merenda, & R. Z. Gold. *Introduction to Bivariate and Multivariate Analysis*. Scott, Foresman, Glenview, IL, 1980. 2, 9

A. H. Nilsson, J. C. Eichstaedt, T. Lomas, A. Schwartz, & O. Kjell. The Cantril ladder elicits thoughts about power and wealth. *Scientific Reports*, 2024. Link. 3, 19

OECD. *OECD Guidelines on Measuring Subjective Well-being*. OECD Publishing, Paris, 2013. Link. 3

S. Pirla, D. Navarro-Martinez, S. Pfattheicher, & J. Quoidbach. Income and boredom: Evidence from 30 countries. *Journal of Economic Psychology*, 2025. Link. 3, 23

N. Powdthavee, R. V. Burkhauser, & J.-E. De Neve. Top incomes and human well-being: Evidence from the Gallup World Poll. *Journal of Economic Psychology*, 2017. Link. 3

E. Proto & A. Rustichini. Life satisfaction, income and personality. *Journal of Economic Psychology*, 2015. Link. 3

M. Rojas, editor. *Handbook of Happiness Research in Latin America*. Springer, Dordrecht, 2016. Link. 3, 12

D. W. Sacks, B. Stevenson, & J. Wolfers. The new stylized facts about income and subjective well-being. *Emotion*, 2012. Link. 2, 15

C. Senik. The French unhappiness puzzle: The cultural dimension of happiness. *Journal of Economic Behavior & Organization*, 2014. Link. 3, 23

B. Stevenson & J. Wolfers. Economic growth and subjective well-being: Reassessing the Easterlin paradox. *Brookings Papers on Economic Activity*, 2008. Link. 2, 3, 15

J. E. Stiglitz, A. Sen, & J.-P. Fitoussi. Report by the commission on the measurement of economic performance and social progress. Technical report, Commission on the Measurement of Economic Performance and Social Progress, Paris, 2009. Link. 25

A. A. Stone & C. Mackie, editors. *Subjective Well-Being: Measuring Happiness, Suffering, and Other Dimensions of Experience*. National Academies Press, Washington, DC, 2013. Link. 3

T. J. VanderWeele, B. R. Johnson, P. T. Bialowolski, et al. The Global Flourishing Study: Study profile and initial results on flourishing. *Nature Mental Health*, 2025. Link. 16

World Bank. Poverty and inequality platform. Line-up estimates of median income or consumption, downloaded on 1 October 2026, 2026. Link. 15

World Bank. World development indicators. GDP, population, household consumption, education, urbanization and age-structure indicators, downloaded from the World Bank API on 27 September–1 October 2026, 2026. Link. 6, 22

World Happiness Report. World happiness report 2026: Data for figure 2.1 (2011–2025). Accessed 27 September 2026, 2026. Link. 5

World Values Survey Association. Rules and procedures for WVSA principal investigators in wave 8, 2023. Link. 22